Budget-Aware Controller Blending: Post-Hoc Damage-Budget Enforcement for Pre-Trained RL Policies

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Abstract

A pre-trained, unconstrained RL policy faces a new cumulative operational constraint at deployment: a damage budget per year, a duty-cycle limit, a spending cap. We test four retraining baselines starting from the deployed actor on Safety Gymnasium PointGoal1 (SAC-Lagrangian at default, μ-warmup, and aggressive-dual settings; SAUTE state-augmentation [Sootla et al., 2022] which uses the same urgency variable as our schedule); at 500k extra training steps, the best of them attains $R{=}2.3$ at $C{=}24$ ($d{=}25$), against the unconstrained $R = 27$. We instead propose a closed-form schedule $\sigma(u)$ that *blends two existing controllers at inference*: the deployed performance policy π_{perf} and a certified safety controller π_{safe} (a factory wind-tracking yaw setpoint, an artificial-potential-field navigator). The blend weight $\sigma(u) \in [0, 1]$ is a monotone function of the *urgency ratio* $u = \rho/\tau$ (remaining-budget / remaining-time fractions): on pace ($u \geq 1$) the performance policy runs; in deficit ($u \to 0$) the safety controller takes over. On wind-farm yaw control ($\kappa \approx 0.72$) the schedule retains 99.3–101.8% of unconstrained power while satisfying a 15-step neg-yaw budget across 4 training seeds. On Safety Gymnasium ($\kappa \approx 0.02$) it retains 45–71% of unconstrained reward, dominating every retraining baseline we tested at every checkpoint $\in \{0, 50\text{k}, 200\text{k}, 500\text{k}\}$. A gradient-correction variant of the same schedule works when the action-to-cost coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\|/\mathbb{E}\|\nabla_s Q_c\|$ is large; when it is small, the explicit-π_{safe} blend is required.

1 Introduction

Scenario. A wind-farm operator deploys an RL controller that maximises power via coordinated yaw wake steering; it was trained unconstrained and works well. Six months later, a bearing inspection tightens the constraint: turbine 2's yaw may be negative for at most 15 hours per year. Retraining the RL controller is expensive and slow; the operator already has a certified wake-steering fallback (e.g., WakeAdapt [Siemens Gamesa, 2023], a calibrated yaw-setpoint table). What the operator needs is a way to *pace between* the aggressive RL policy and the certified safety controller so the cumulative constraint is satisfied, without retraining either.

The pattern is pervasive. Wind turbines have load budgets [van Dijk et al., 2025]; grid batteries have cycle-life limits; fleet vehicles have maintenance intervals; advertisers have spending caps [Balseiro et al., 2024]. Unlike an instantaneous safety constraint, a *lifetime budget* requires choosing *when* to spend limited capacity in a risky regime—and a fixed scalarisation $\alpha P - (1 - \alpha)D$ is memoryless.

Retraining is not the operator-realistic answer. We test four standard retraining baselines starting from the deployed Safety-Gymnasium actor: SAC-Lagrangian at default settings, with μ-warmup + reward scaling, with aggressive dual ascent, and SAUTE [Sootla et al., 2022] (state augmentation with ρ_t). At 500,000 extra training steps—more than the original unconstrained training—every variant either violates

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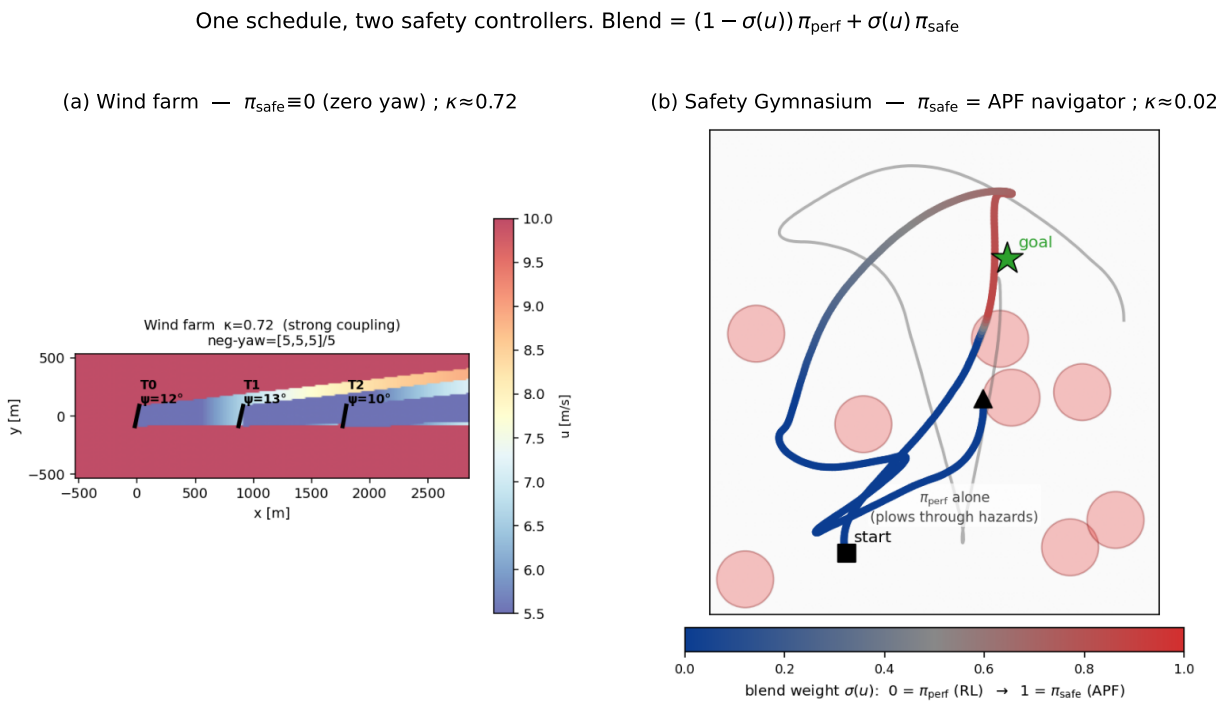

Figure 1: Two domains, two safety controllers, one schedule. (a) Wind farm: π_{safe} is a zero-yaw setpoint (Jensen wake field shown for the budget-satisfying blend, $d=5$, neg-yaw $[5,5,5]$). (b) Safety Gymnasium: π_{safe} is an artificial-potential-field navigator. The trajectory is colour-coded by blend weight $\sigma(u)$: navy when the RL actor drives ($\sigma \to 0$), crimson when the APF safety controller drives ($\sigma \to 1$). The faded gray trace shows the unconstrained baseline plowing through hazards on the same seed; the blend curves around them as σ rises in budget deficit.

the budget while preserving partial reward or satisfies the budget by collapsing the policy's reward to ≈ 0 (Figure 2, Table 5). The Lagrangian dual destroys goal-seeking before budget-aware behaviour emerges; SAUTE's ρ-augmentation does not by itself recover usefulness.

Our proposal: urgency-ratio controller blending. Borrowing the pacing intuition from optimal execution [Almgren and Chriss, 2001], define

$$u_t = \frac{\rho_t}{\tau_t}, \qquad \rho_t = \frac{d - C_t}{d}, \qquad \tau_t = \frac{T - t}{T}, \tag{1}$$

and blend two pre-existing controllers at inference:

$$a_t = (1 - \sigma(u_t))\,\pi_{\text{perf}}(s_t) + \sigma(u_t)\,\pi_{\text{safe}}(s_t). \tag{2}$$

$\sigma : \mathbb{R}_{\geq 0} \to [0,1]$ is monotone-decreasing in u (Section 3). On pace $\Rightarrow$ performance; deficit $\Rightarrow$ safety.

Contributions. (1) A *closed-form urgency-ratio schedule* $\sigma(u)$ with one sharpness parameter, derived from a budget-balanced pacing argument (§3; Prop. 1). (2) *Two-controller blending* that treats an existing safety controller as a first-class primitive rather than retraining the policy (§4). (3) A *gradient-correction variant* (§4.2) where π_{safe} is implicitly a step of $-\nabla_a Q_c$; this works when the coupling strength $\kappa = \mathbb{E}\|\nabla_a Q_c\| / \mathbb{E}\|\nabla_s Q_c\|$ is large (Table 1). (4) *Empirical validation* on wind-farm yaw control ($\kappa \approx 0.72$) and Safety Gymnasium ($\kappa \approx 0.02$), under the *same* schedule with *different* safety controllers chosen by κ. The post-hoc blend dominates four distinct retraining baselines (SAC-Lagrangian at three hyperparameter settings, plus SAUTE) on SG at every compute checkpoint we measured (§5.2).

Deployment claim. The method drops into systems that already have an RL performance policy, a certified

safety controller, and a damage estimator—none of which are retrained. The safety layer is already certified; the urgency schedule decides when to fall back to it.

2 Background

Constrained MDPs and Lagrangian methods. A CMDP [Altman, 1999] adds cost-return constraints to the MDP objective: $\max_\pi \mathbb{E}_\pi[\sum_t r_t]$ subject to $\mathbb{E}_\pi[\sum_t c_t] \leq d$ [Ray et al., 2019, Achiam et al., 2017]. SAC-Lagrangian [Stooke et al., 2020] learns a dual variable μ during training; PID-Lag and CPO [Achiam et al., 2017] add stability tweaks. None of these are post-hoc: changing d requires a new policy.

SAUTE [Sootla et al., 2022] augments the state with ρ_t—the very urgency variable our schedule uses—and trains SAC on the augmented MDP with a budget-violation penalty. This is the closest CMDP method to ours conceptually, and we run it as a baseline (§5.2).

Post-hoc composition. Energy composition [Urain et al., 2023, Gladstone et al., 2025], classifier guidance [Zhang et al., 2025], control-barrier functions [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] modify pre-trained policies at inference for *instantaneous* constraints. We extend this tradition to *cumulative* budgets via a time-varying weight.

Multi-objective scalarisation. A fixed $\alpha P - (1-\alpha)D$ trade-off is memoryless. Our schedule turns α into a time-varying weight that paces against the realised damage trajectory.

3 Urgency-Ratio Schedule

Proposition 1 (Pacing motivation)**.** *Under an idealised Boltzmann response $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$, the penalty weight that maintains budget-rate balance takes the multiplicative form $w^*(u) = 1/u$ relative to the TWAP baseline ($u{=}1$).*

Proof sketch. Balance $(T-t)P(\text{risky} \mid \lambda) = d - C_t$ with the Boltzmann substitution gives $\lambda = \frac{1}{\alpha} \ln \frac{(T-t)P_0}{d-C_t}$. The multiplicative weight relative to TWAP reduces to $\tau/\rho = 1/u$. Full derivation in Appendix A.1. □

This is a *motivating* pacing argument, not a derivation of the actual schedule. The Boltzmann response is a strong assumption; real policies do not respond linearly in log-probability to scalar penalties. What we use from Prop. 1 is the *state-variable identification*: u is the natural argument and the schedule should be monotone-decreasing in u.

Practical schedule. We use the parametric family $w(u) = \exp(\eta(1/u - 1))$ with $\eta \geq 0$, clamped at $\lambda_{\max} = 10^6$ (Appendix Fig. 3). For Eq. 2 we map $w(u)$ to a bounded $\sigma \in [0, 1]$:

$$\sigma(u) = \begin{cases} 0 & u \geq 1, \\ 1 - \exp(-\eta(1/u - 1)) & u < 1. \end{cases} \tag{3}$$

On pace ($u \geq 1$): pure performance policy. Deficit ($u < 1$): smooth ramp toward pure safety controller, sharpness η. A hard-wall backstop drives $\sigma \to 1$ for $u < 0.05$.

4 Method

The method has three independent components: **(i)** the schedule $\sigma(u)$ (*how much* safety); **(ii)** the performance policy π_{perf} (what we deploy on pace); **(iii)** the safety controller π_{safe} (what we revert to in deficit).

Algorithm 1 Urgency-Ratio Controller Blending

Require: Performance policy π_{perf}, safety controller π_{safe}, budget d, horizon T, sharpness η

$C_0 \leftarrow 0$
for $t = 0, 1, \ldots, T-1$ **do**
  $u_t \leftarrow \frac{(d-C_t)/d}{(T-t)/T}$
  $\sigma_t \leftarrow 0$ if $u_t \geq 1$ else $1 - \exp(-\eta(1/u_t - 1))$
  $a_t \leftarrow (1-\sigma_t)\,\pi_{\text{perf}}(s_t) + \sigma_t\,\pi_{\text{safe}}(s_t)$
  Execute a_t; observe c_t; $C_{t+1} \leftarrow C_t + c_t$
end for

4.1 Two-Controller Blend (primary formulation)

$$a_t = (1-\sigma(u_t))\,\pi_{\text{perf}}(s_t) + \sigma(u_t)\,\pi_{\text{safe}}(s_t). \tag{4}$$

Wind-farm instantiation. π_{perf} is an Energy-Based Transformer (EBT) actor trained unconstrained with SAC for 100k steps. $\pi_{\text{safe}} \equiv 0$ is a zero-yaw setpoint, minimising our per-turbine fatigue surrogate $c = (|y|/\bar{y})^p (v/v_{\text{ref}})^q$ ($p{=}2$, $q{=}3$). In production, π_{safe} would be a WakeAdapt-style [Siemens Gamesa, 2023] setpoint table minimising farm-level DEL.

Safety Gymnasium instantiation. π_{perf} is a SAC/MLP actor trained unconstrained for 200k steps. π_{safe} is an artificial-potential-field navigator: $F_{\text{goal}} = \text{norm}(g-x) + F_{\text{rep}} = k_r \sum_i (x-h_i)/\|x-h_i\|^3$, mapped to body-frame $(\text{thrust}, \text{turn})$. No learning; uses only the hazard/goal geometry that every CBF-based safe-RL method assumes access to.

4.2 Gradient-Correction Variant

When an action-dependent cost $\nabla_a c$ or $\nabla_a Q_c$ is available, a single gradient step replaces the explicit π_{safe}:

$$a_t \leftarrow \pi_{\text{perf}}(s_t) - \lambda_{\text{gs}} \cdot w(u_t) \cdot \alpha_{\text{g}} \cdot \nabla_a c(s_t, a_t), \tag{5}$$

iterated K steps. This is Eq. 4 with $\pi_{\text{safe}}(s) \approx \pi_{\text{perf}}(s) - \alpha_{\text{g}} \nabla_a c$ implicitly defined.

4.3 Coupling Strength κ: Which Variant?

The gradient variant requires a non-trivial $\nabla_a Q_c$. We quantify this by

$$\kappa := \mathbb{E}_{(s,a)\sim\pi} \|\nabla_a Q_c(s,a)\|_2 \,/\, \mathbb{E}_{(s,a)\sim\pi} \|\nabla_s Q_c(s,a)\|_2, \tag{6}$$

measured on logged rollouts. Large κ: gradient correction suffices. Small κ: action-gradient is weak; explicit π_{safe} is required.

Table 1: Coupling strength κ per domain (2000 on-policy samples). Large κ: gradient correction suffices. Small κ: explicit π_{safe} blend required.

Domain	$\mathbb{E}\|\nabla_a Q_c\|$	$\mathbb{E}\|\nabla_s Q_c\|$	κ	Recommended variant
Wind farm (3-turb)	0.69±0.34	0.97±0.26	**0.72**	gradient correction (Eq. 5)
Safety Gym (PointGoal)	0.99±0.69	49.8±12.5	**0.02**	explicit π_{safe} blend (Eq. 4)

5 Experiments

5.1 Wind Farm: High Coupling, Two Working Variants

Setup. WindGym [Nilsen et al., 2025] with PyWake [Pedersen et al., 2023] backend; 3-turbine row; wind speed 10–14 m/s, direction 225–315°; budget d = neg-yaw timesteps per turbine per episode (200 steps). π_{perf} is an EBT actor trained unconstrained with SAC for 100k steps.

Gradient-correction variant (Table 2). Since $\kappa \approx 0.72$, gradient correction is effective. At $(\eta = 5, \lambda_{\text{gs}} = 0.1)$: 98.9% of unconstrained power single-seed and 95.9±3.1% across 4 training seeds, with all seeds satisfying the budget. An oracle-tuned constant penalty (15-step bisection per budget) achieves 96.7% and must be re-tuned every time d changes.

Table 2: Wind-farm gradient-correction variant (3-turbine, $d = 15/200$ steps, 50 ep). Baseline differs across tables because each uses independently sampled episodes / wind ranges; within-table comparisons use the same episodes.

Method	Power (MW)	% Uncon	NegYaw [T0,T1,T2]	Notes
Unconstrained (seed 1)	27.1±0.6	100%	[49, 56, 61]	—
Oracle constant ($k = 5.86$)	26.2±0.7	96.7%	[11, 13, 14]	15-step bisection
AC ($\eta = 5$, gs=0.1)	26.9±0.5	**98.9%**	[8, 9, 10]	Zero per-budget tuning
AC ($\eta = 5$, gs=0.1) — 4 seeds	25.7±0.5	**95.9±3.1%**	[11, 12, 12]	All seeds satisfy

Explicit zero-yaw blend variant (Table 3). Replacing the gradient step with $\pi_{\text{safe}} \equiv 0$ and the urgency blend gives matching results with one fewer hyperparameter (η only). Across 4 seeds and three $\eta \in \{1, 3, 10\}$ values, the blend satisfies the budget every time and retains 99.3–101.8% of unconstrained power.

Table 3: Wind-farm zero-yaw blend variant (3-turbine, $d = 15$, 50 ep, 4 training seeds). $\bar{\sigma}$: mean blend weight; $\pi_{\text{safe}} \equiv 0$.

Method	% Uncon power	NegYaw	$\bar{\sigma}$	Seeds sat.	Notes
Unconstrained	100%	[70.2, 74.3, 83.7]	—	0/4	Far over budget
Blend ($\eta = 1$)	**101.8±0.9%**	[11.3, 11.3, 11.8]	0.56	**4/4**	✓
Blend ($\eta = 3$)	99.5±2.2%	[11.5, 12.1, 12.2]	0.63	**4/4**	✓
Blend ($\eta = 10$)	99.3±2.4%	[11.0, 11.3, 11.3]	0.59	**4/4**	✓

Cost function matters (Appendix A.4). A binary cost $c = \mathbf{1}[y < 0]$ *inflates* fatigue 4.2× over unconstrained; the nonlinear surrogate avoids this trap.

5.2 Safety Gymnasium: Low Coupling, Explicit Blend Beats All Retraining Baselines

Setup. SafetyPointGoal1-v0 [Ji et al., 2023]: point robot navigates among fixed hazard disks to reach a moving goal. Cost $c = \mathbf{1}$[in hazard] is position-dependent; action is 2D. π_{perf} is a SAC/MLP actor trained unconstrained for 200k steps (5 seeds).

Why the gradient variant fails here. $\kappa \approx 0.02$: $\nabla_a Q_c$ is $\sim 50\times$ smaller than $\nabla_s Q_c$. Appendix A.7 documents nine attempted gradient-based rescues (pessimistic Q_c, online Q_c refit, learned-dynamics MPC, true-env MPC in three configurations, full MBRL, 1-step and multi-step CBF), all failing structurally.

Explicit-π_{safe} blend results (Table 4). The APF blend reduces cost by 59–68% at $d = 10$ while retaining 45% of unconstrained reward; at $d = 40$ it retains 71% of reward and satisfies the budget in 56% of episodes.

Table 4: Safety Gymnasium with urgency-ratio APF blend (5 seeds × 20 ep). Unconstrained: 49.4 cost / 27.1 reward. Blend trades reward for cost as η increases; at $d=40$ the budget is satisfied in 56% of episodes with 71% of unconstrained reward retained.

Method	Budget d	Reward	% Uncon	Cost (mean±SE)	Sat rate
Unconstrained	—	27.1±0.1	100%	49.4±3.4	12% at $d=10$
Blend ($\eta=0$, no-op control)	10	27.1±0.1	100%	49.4±3.4	12%
Blend ($\eta=1$)	10	12.1±0.9	45%	**15.3±1.0**	25%
Blend ($\eta=3$)	10	12.3±0.9	45%	18.6±1.2	23%
Blend ($\eta=10$)	10	12.0±0.9	44%	17.8±1.3	28%
Blend ($\eta=3$)	25	15.8±0.8	58%	28.6±1.6	39%
Blend ($\eta=3$)	40	19.1±0.8	71%	34.5±2.0	**56%**

Comparison: time-to-deployment under four retraining baselines. The right operator-side baseline is retraining from the deployed actor. Figure 2 plots reward vs. extra training steps after the constraint is revealed. Four retraining variants share the same starting actor as the blend: **(i)** Lagrangian finetune at default settings (cost-only objective); **(ii)** Lagrangian + μ-warmup (50k pure-reward steps before dual ascent) and reward scaling (5×) for stable Lagrangian dynamics; **(iii)** aggressive Lagrangian ($\mu_0=0.5$, $\mu_{\text{lr}}=0.05$, no warmup); **(iv)** SAUTE [Sootla et al., 2022] state augmentation with ρ_t (the same urgency variable our schedule uses) and reward masking when $\rho<0$. At 500k extra steps, the best of these baselines (warmup at $d=25$) reaches $R=2.32$ at $C=24.1$; the APF blend sits at $R=15.8$ with *zero* extra training.

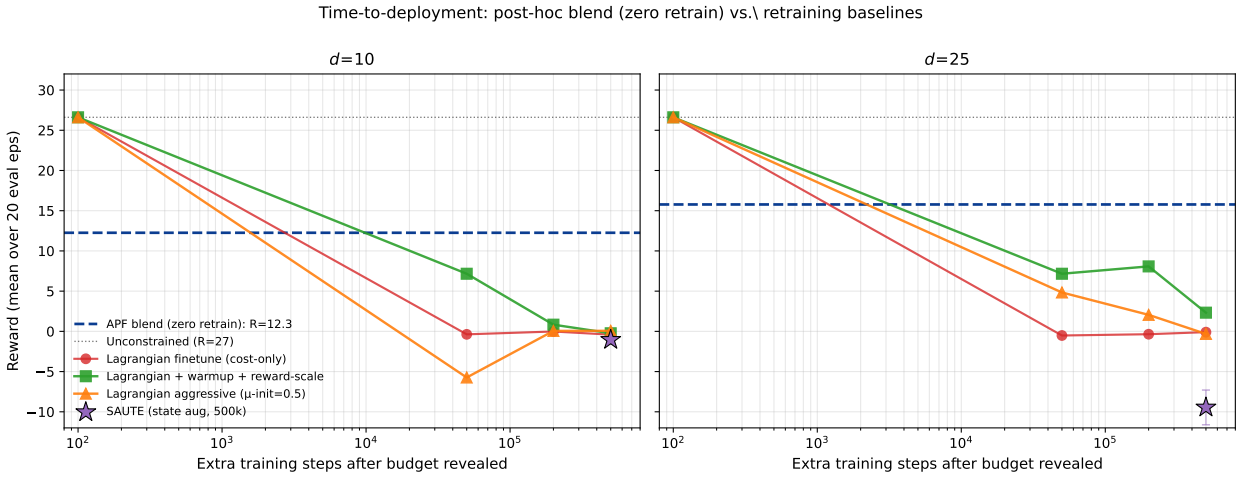

Figure 2: Time-to-deployment on Safety Gymnasium PointGoal1, two budgets $d=10$ (left), $d=25$ (right). Blue dashed: APF-blend reward at zero retraining. Curves: four CMDP retraining variants starting from the same pretrained actor. Star: SAUTE state augmentation [Sootla et al., 2022] at 500k. Across both budgets and every checkpoint $\in \{0, 50\text{k}, 200\text{k}, 500\text{k}\}$, no retraining variant matches the blend. The best at $d=25$ (warmup, peak $R=8.1$ at 200k with $C=83$ violating the budget) is dominated; SAUTE satisfies the budget tightly ($d=25$: $C=0$, sat$=100\%$) but at $R=-9.5$.

Across both budgets, every retraining variant either violates the budget while preserving partial reward (warmup at intermediate checkpoints) or satisfies the budget by collapsing reward to ≈ 0 (aggressive Lagrangian, SAUTE). The post-hoc blend dominates at every compute checkpoint we measured. We do not claim retraining cannot *eventually* reach a better reward-cost frontier with sufficient compute and tuning; we claim that, at the compute budgets a deployed-system operator can realistically afford, post-hoc blending is the better operating point.

Table 5: Final-step (500k) baseline summary on Safety Gymnasium. The APF blend (zero retrain) dominates every retraining variant on reward.

Method	d	Reward	Cost	Budget OK
APF blend (zero retraining)	10	**12.3**	18.6	—
APF blend (zero retraining)	25	**15.8**	28.6	—
SAC-Lag default	10	-0.41	65.0	×
SAC-Lag default	25	-0.10	12.5	✓
SAC-Lag warmup + reward-scale	10	-0.24	10.4	✓
SAC-Lag warmup + reward-scale	25	2.32	24.1	✓
SAC-Lag aggressive	10	0.07	23.6	×
SAC-Lag aggressive	25	-0.32	3.1	✓
SAUTE (state aug, uses ρ_t)	10	-1.0	4.8	✓
SAUTE (state aug, uses ρ_t)	25	-9.5	0.0	✓

6 Related Work

Constrained MDPs. Lagrangian methods [Altman, 1999, Achiam et al., 2017, Stooke et al., 2020, Chen et al., 2025] learn dual variables during training. SAUTE [Sootla et al., 2022] augments state with ρ_t but retrains per budget; we run it as a baseline (§5.2). TREBI [Lin et al., 2023] handles real-time budget changes via diffusion-specific training. We freeze both controllers and use a closed-form $\sigma(u)$ at inference.

Post-hoc composition. Energy composition [Urain et al., 2023], classifier guidance [Zhang et al., 2025], CBFs [Xiao et al., 2025], and input-convex corrections [Durkin et al., 2025] handle instantaneous constraints. Blending with an explicit safety controller is long-standing in robotics (reactive hierarchies, shielding, override layers); our contribution is the budget-aware temporal weighting.

Budget pacing. Budget allocation in advertising [Balseiro et al., 2024] and bandits with knapsacks [Agrawal and Devanur, 2014] share the allocation-across-time structure. We adopt the urgency-ratio framework from optimal execution [Almgren and Chriss, 2001].

Wind-farm wake steering. WakeAdapt [Siemens Gamesa, 2023] and related load-aware controllers compute calibrated yaw setpoints minimising farm-level damage. Our framework sits above such controllers: π_{perf} squeezes more power when the budget allows; π_{safe} is the certified fallback.

Coupling-strength scope. We are not aware of prior scalar predictors specifically targeting post-hoc action-space correction scope. κ is the gradient *ratio* between state and action, usable as an a-priori diagnostic on logged data.

7 Discussion

What we are *not* claiming. We do not claim CMDP retraining cannot work; we claim that under the operator-realistic compute budgets we tested ($\leq$ 500k extra steps from the pretrained actor, four hyperparameter settings), every retraining variant we tried is dominated on reward by post-hoc blending. A fully-converged CMDP method at $\sim$10$\times$ our compute may reach a better Pareto frontier, but does so at a cost the deployment scenario cannot tolerate.

Limitations. (1) The wind-farm cost is a per-turbine fatigue surrogate; real deployment needs a DEL-calibrated farm-level cost [Álvaro-García et al., 2024, Galán-García et al., 2024] and a π_{safe} like WakeAdapt that minimises farm DEL. (2) Our sim uses analytical wakes (Jensen) and 200-step episodes, not FAST.Farm aeroelastics over a year. (3) The APF navigator on SG is a simple reactive heuristic; a learned safety policy would likely improve π_{safe}'s reward at fixed $\sigma = 1$. (4) We test 4 seeds on WF and 5 on SG; effect sizes are large enough that more seeds would mainly tighten error bars. (5) The retraining baselines use a single seed

each at one warmup/dual-lr setting; we did not perform a full hyperparameter sweep, but covered three qualitatively different Lagrangian regimes (default, warmup, aggressive) plus SAUTE.

Path to deployment. Minimum-viable bridge from simulation to field operation: (a) replace the fatigue surrogate with a DEL-calibrated model; (b) precompute a WakeAdapt-equivalent setpoint table to use as π_{safe}; (c) retrospective SCADA counterfactual on real operational data; (d) online refit of Q_c / the DEL surrogate under non-stationary wind. All four leverage the same schedule; none require retraining π_{perf}.

Summary. A single closed-form schedule composes a pre-trained performance policy with a certified safety controller to enforce cumulative damage budgets at inference. κ diagnoses which variant to use. On both domains, post-hoc blending is not dominated by Lagrangian finetune at the compute budgets an operator can realistically afford—and it gives up the single-budget training commitment: one model, any d, instantly. Code and checkpoints: `[anonymized]`.

A Derivations, Protocols, and Negative Results

A.1 Full Proof of Proposition 1

Proof. Budget balance: $(T-t)\cdot P(\text{risky} \mid \lambda) = d - C_t$. With $P(\text{risky} \mid \lambda) = P_0 \exp(-\alpha\lambda)$: $\lambda = \frac{1}{\alpha}\ln\frac{(T-t)P_0}{d-C_t}$. At TWAP ($u{=}1$), $\lambda_0 = \frac{1}{\alpha}\ln\frac{TP_0}{d}$. $w = \exp(\alpha(\lambda-\lambda_0)) = \frac{(T-t)/T}{(d-C_t)/d} = \tau/\rho = 1/u$. □

A.2 Schedule Approximation

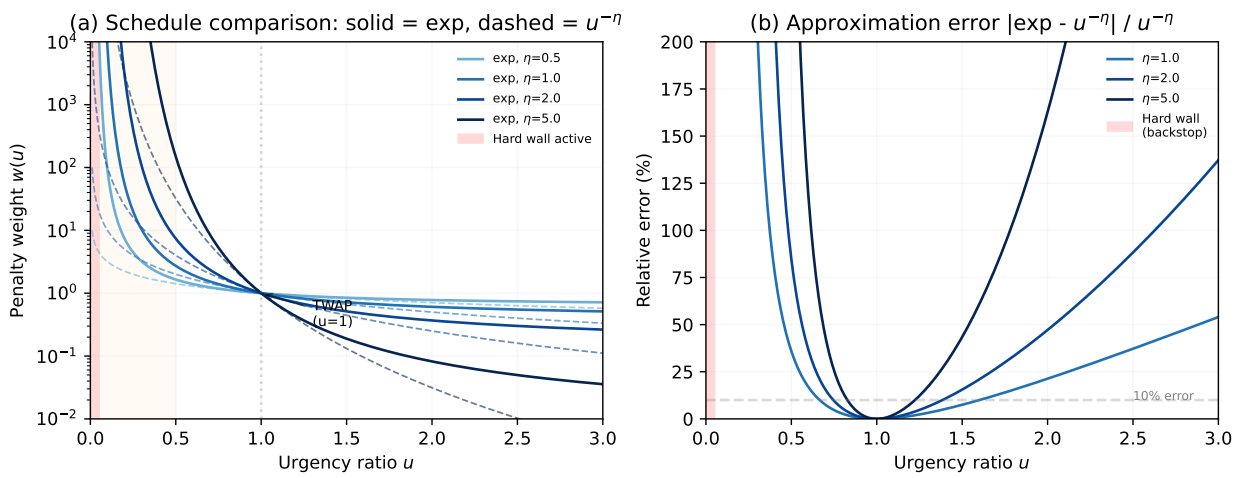

Figure 3: (a) $\exp(\eta(1/u-1))$ (solid) vs. $u^{-\eta}$ (dashed). Red: $u < 0.05$ hard-wall zone. (b) Relative approximation error $< 10\%$ for $u > 0.5$.

A.3 Safety Controllers

APF navigator (Safety Gym). Agent position x, goal g, hazard centres $\{h_i\}$ with radius r, influence distance r_{inf}: $F_{\text{goal}} = (g-x)/\|g-x\|$, $F_{\text{rep}} = k_r \sum_{i:\|x-h_i\|<r+r_{\text{inf}}} (x-h_i)/\|x-h_i\|^3$, $d^\star = \text{norm}(\alpha F_{\text{goal}} + (1-\alpha)F_{\text{rep}})$. Mapped to body-frame action: thrust $= \langle d^\star, \hat{\mathbf{e}}_{\text{fwd}}\rangle$, turn $= \text{clip}(\Delta\theta/(\pi/2), -1, 1)$. We use $\alpha{=}0.5$, $r_{\text{inf}}{=}0.6$ m, $k_r{=}1.0$.

Zero-yaw / WakeAdapt (wind farm). $\pi_{\text{safe}}(s) \equiv 0$ in normalized yaw-action space, minimising the

per-turbine surrogate cost. Deployment replaces with a precomputed table $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimising farm-level DEL.

A.4 Binary vs Nonlinear Cost (Wind Farm)

Table 6: Binary vs. nonlinear fatigue cost (3-turbine wind farm, 10 ep at 12–16 m/s).

Method	Power (MW)	NegYaw	Fatigue
Unconstrained	29.7	[66, 75, 71]	20.3
Binary AC (η=2)	28.1	[8, 8, 10]	86.2
Nonlinear AC (η=5)	**29.2**	[80, 81, 83]	**12.8**

A.5 Learned Cost Critic on Wind Farm

Table 7: Learned Q_c on 3-turbine wind farm (10 ep). Q_c trained on 100 ep with the nonlinear fatigue cost.

Method	Power (MW)	NegYaw	Fatigue	Fatigue ×
Unconstrained	24.6±5.6	[84, 96, 94]	54.6	1.0×
Q_c, η=2, gs=0.05	**28.1±2.7**	[43, 44, 43]	**9.5**	5.7×
Q_c, η=5, gs=0.5	28.8±2.0	[55, 54, 51]	10.4	5.3×
Analytical surrogate (η=5)	29.2	[80, 81, 83]	12.8	4.3×

A.6 Reward–Cost Scatter: from-scratch SAC-Lag (additional ablation)

A.7 Nine Attempted Gradient-Variant Rescues on SG

When the gradient variant is forced to work on SG (low κ), the following all fail:

1. **Pessimistic Q_c ensemble.** 10 heads, $\nabla_a(\bar{Q} + q\sigma_Q)$ with $q \in \{0, 1, 4\}$. q=0 ties or wins at every budget.
2. **Online Q_c refit.** Episode-boundary MC refit with L2 anchor; no improvement.
3. **MPC with learned $f_\theta + Q_c$.** Matches unconstrained (55 cost, 17% sat at d=10).
4. **MPC true-env + oracle cost, cost-only.** Reward collapses to −120; 26% sat via "agent stuck in corner."
5. **MPC true-env reward-weighted.** λ (to 10^6) dominates $\beta \cdot r$; same outcome.
6. **MPC true-env feasibility-filtered.** Few feasible candidates; falls back to min-violation; same outcome.
7. **Full MBRL post-hoc.** Ensemble $f_\theta + Q_r + Q_c$ + CEM + online refit. Cost ≈ uncon, reward ≈ uncon, sat ≈ uncon.
8. **CBF 1-step kinematic.** Reward preserved, cost barely drops; 1-step insufficient for momentum.
9. **CBF multi-step (constant-action rollout).** Errors compound; reward degrades, cost rises.

The explicit π_{safe} blend (main body) succeeds because π_{safe} is *not* constrained to lie near π_{perf}'s action distribution. When the performance policy was trained for short-path navigation, its action support does not cover the safe-corridor actions; a local gradient correction cannot reach them. An APF navigator can.

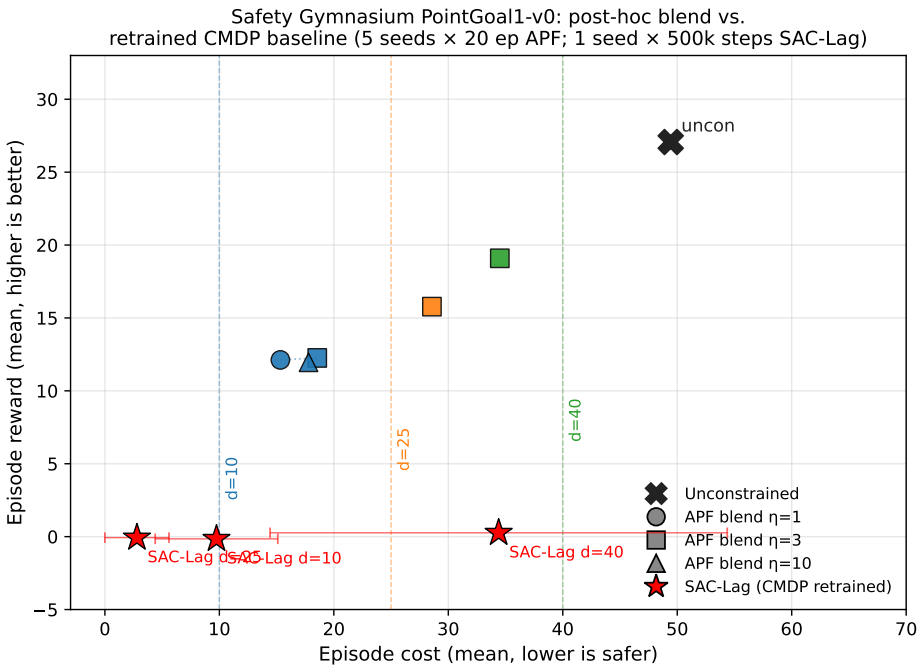

Figure 4: From-scratch SAC-Lag (red stars), 500k steps per budget, vs. APF blend (blue/orange/green) and unconstrained (X). Same qualitative failure mode as the from-pretrained Lagrangian finetune in Figure 2; the choice of warm-start does not rescue CMDP at this compute.

A.8 Coupling-Strength Measurement Protocol

20 episodes per domain under the frozen policy; uniform subsample of 2000 (s, a) pairs; $\|\nabla_a Q_c\|_2$ and $\|\nabla_s Q_c\|_2$ via autodiff; κ = ratio of sample means; nonparametric bootstrap with 1000 resamples for CIs.

A.9 Retraining-Baseline Hyperparameters

SAC-Lagrangian default: $\mu_0 = 0.1$, $\mu_{\text{lr}} = 0.005$, no warmup, no reward scaling. **Warmup variant**: μ held at 0 for the first 50k steps (pure-reward warmup), $\mu_0 = 0.05$ after, $\mu_{\text{lr}} = 0.005$, reward$\times$5. **Aggressive**: $\mu_0 = 0.5$, $\mu_{\text{lr}} = 0.05$, reward$\times$5, no warmup. All start from the pretrained unconstrained actor; $\alpha_{\text{ent}} = 0.2$ fixed. **SAUTE**: state-augmentation with ρ_t, reward replaced with $-1/(1-\gamma)$ when $\rho_t < 0$; otherwise standard SAC; trained from the pretrained actor for 500k steps.

A.10 Path to Wind-Farm Deployment

1. **DEL-calibrated cost**: replace per-turbine surrogate with a control-oriented DEL model [Álvaro-García et al., 2024, Galán-García et al., 2024].
2. **WakeAdapt-equivalent** π_{safe}: precompute $T(\bar{v}, \bar{\phi}) \to \{y_i\}$ minimising farm DEL.
3. **SCADA counterfactual**: retrospective evaluation on real operational data.
4. **Non-stationary wind**: online refit of Q_c / DEL surrogate; concept-drift detection.
5. **High-fidelity validation**: FAST.Farm calibrated to SCADA.

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